Agenda

→ Intro to LLD

→ Why LLD is important

→ Structure of LLD module

→ FAQ's

What is LLD?

↓ low level design.

↳ opposite of LLD → HLD (High level design).

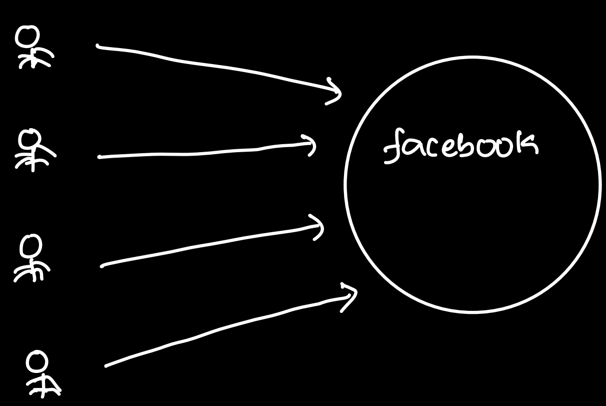
general outline

not too much into detail

Superfecial

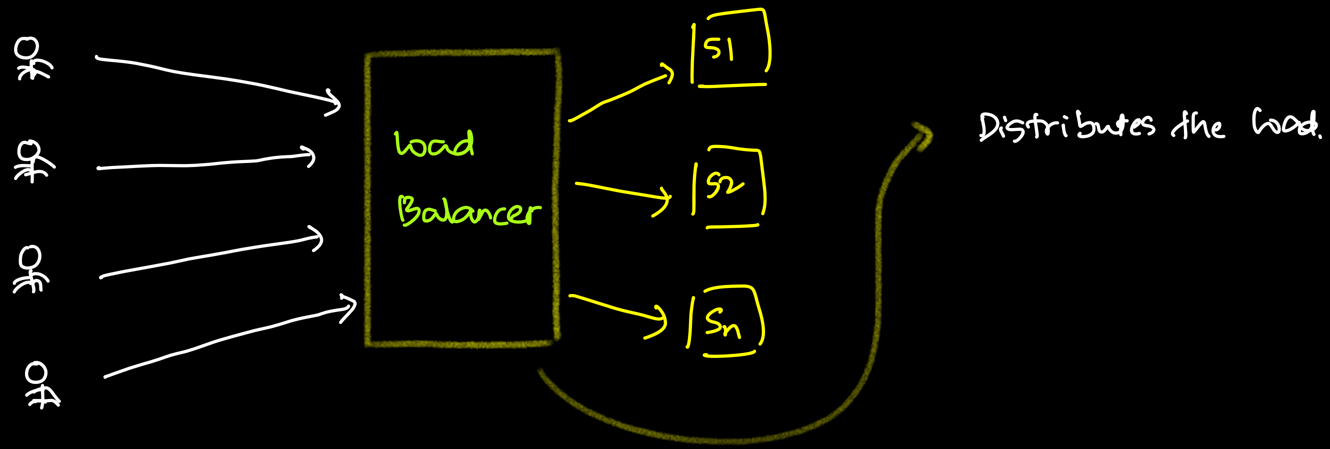
overview

ex: How facebook works? -



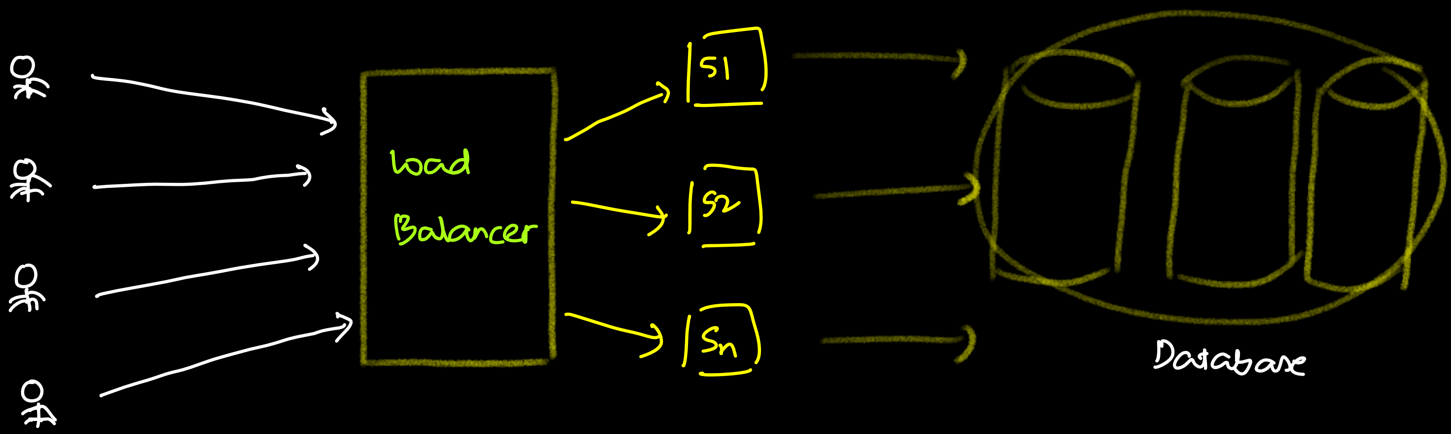
Can one server handle all requests? .. NO..

FB has Billions of requests.

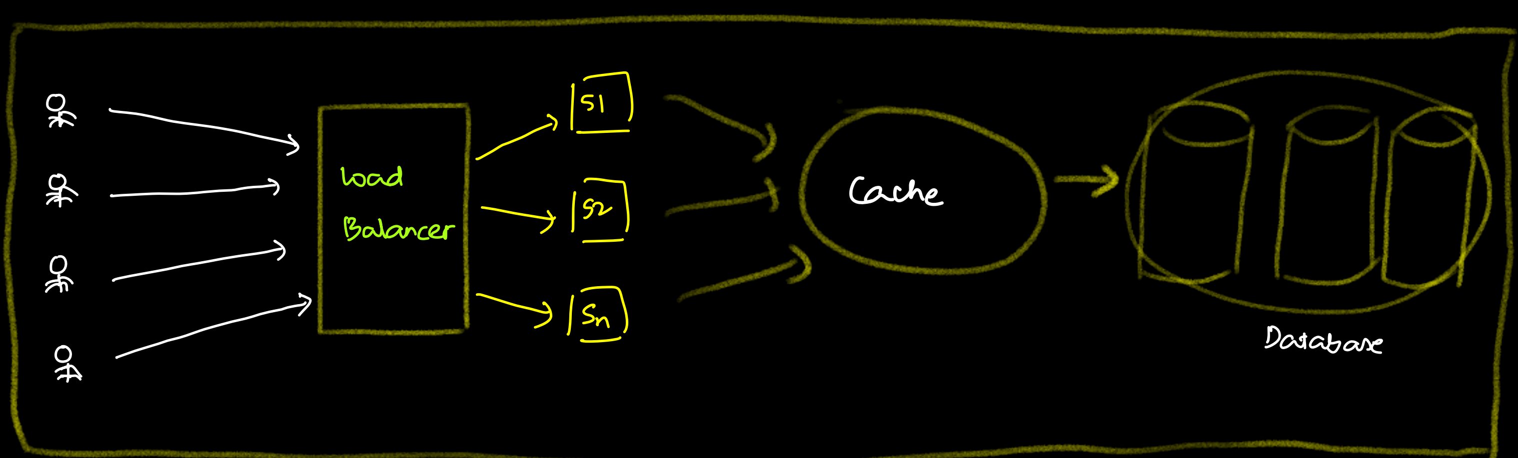


Q: Where's all the data stored? -

ex: user profile ⇒ database.



Not all the requests needs to hit the database. -



Defn:

Deals with different infrastructure levels (DB, Server, load balancer, cache) that'll work together for the efficient working of the sysrem.

LLD

* deals with the details of code that is necessary for the components to interact with.

* Implementation of the system.

  [classes, attributes, how classes will interact].

LLD is also called as "Object Oriented Design".

Why LLD?

Daily job

What % of time goes into writing code?..

12% of the time is gone into writing new code.

Rest of the time -- (that helps the company)

* Traffic
* Gossips/chit chat
* LinkedIn / insta
* Breaks / lunch / tea
* Gyms

* Meetings / scrum
* Documentation
* code review
* Debug code
* fix Bugs
* Interviews

* Testing
* Learning

LLD helps in these

It helps you write code

* Understandable
* Reusable
* Extensible
* Maintainable
* Modular

Interviews

SDE 1, 2, 3 => atleast 1 round of LLD.

Theory Questions

↓

Tier 3 -> 2 companies.

class diagrams

Machine coding

↓

More common in Product based companies

Structure of LLD module

1. OOPS -> [4]
2. Concurrency 4 Threads -> [4-5]
3. Advanced language concepts -> [3]

} LLD - 1

1. SOLID principles -> [2]
2. Design patterns -> [6-7 classes]
3. UML diagram -> [1]

} LLD-2

1. How to approach LLD (pen) → [1]
2. Tic Tac Toe → [3-4]
3. Parking Lot → [3-4]
4. Book My Show → [3-4]
5. Splitwise → [3]

$\left.\right\}$ LLD-3

28th October 2024.

FAQ's

1. Why java?..
2. Assignments
3. CS Fundamentals

* DBMS (SQL)
* OS [LLD module] [HLD]
* Computer Networks [Project Module] [HLD]

4. Job Readiness

SDE-1 → DSA + SQL + LLD

SDE-2 & above → DSA + SQL + LLD + HLD + Project

How to Revise DSA

Arrays

1. question name
2. ______
3. ______

write code for any 1 question

hint 1

hint 2

Github link

Linked list

Trees